ESE 4105 One-Shot Notes

Fuel Combustion and IC Engines · KUET · past papers 2020, 2021, 2022, 2023, 2024 · answers from the course slides

One-shot notes: ESE 4105 Fuel Combustion and IC Engines

Fuel properties

Octane = % iso-octane in iso-octane + n-heptane matching knock (resist auto-ignition; SI). **Cetane** = % n-hexadecane in hexadecane + 1-methylnaphthalene matching delay (ready to ignite; CI). ON 85 = knocks like 85/15 blend; CN 60 = good diesel.

API = 141.5/SG − 131.5 (hydrometer; light > 31.1, heavy < 22.3). **RVP**: 38 °C, air chamber 4× fuel chamber, volatility (start vs vapour lock). **Aniline point**: aniline-oil miscibility temperature, low = aromatic = poor diesel. **Diesel index** = AP(°F) × API / 100. Flash point, fire point (5 s), pour, cloud point.

Analysis and CV

Proximate: moisture 108-110 °C, VM 950 °C 7 min covered, ash 725 °C, FC by difference. **Ultimate**: C, H (KOH/$CaCl_2$), N (Kjeldahl), S ($BaSO_4$), O by difference.

HCV = 33 800C + 144 000(H − O/8) + 9 270S kJ/kg. LCV = HCV − 9H × 2442. Stoich air = (2.67C + 8H + S − O)/0.23 kg/kg.

Fuels: classes, storage, alternatives

Primary (wood, coal, crude, NG) vs secondary (coke, petrol, LPG, producer gas); solid/liquid/gas; fossil, by-product, chemical, nuclear.

Storage: solid = self-heating + dust; liquid = vapour, flash point, bunds, earthing; gas = leaks, pressure, odorant, ventilation.

Ethanol ON ~108, E10/E85, LCV 2/3. Biodiesel (FAME, transesterification) higher CN, less CO/HC/soot, bit more NOx. Methanol: half CV, toxic, corrosive, cold start, formaldehyde. H_2: SIT ~585 °C → dual fuel with diesel pilot / glow plug; 4-75 % flammable; only NOx.

Refining: distillation 300-350 °C column 20-30 m, cut every ~50 °C; cracking (heavy → light); reforming (Pt, 480-540 °C, naphthenes → aromatics, raises ON).

Combustion and A/F

Fire triangle: fuel + O_2 + ignition temperature. Types: complete, incomplete, rapid, spontaneous, explosion; premixed vs diffusion.

$\varphi = (F/A)/(F/A)_s = 1/\lambda$. Stoich A/F: petrol 14.7, diesel 14.5, CH_4 17.2. Best power ~12.5-13 (rich), best economy ~16-17 (lean).

Producer gas (H_2 25, CO 15, CH_4 5, CO_2 15, N_2 40): O_2 = 0.125 + 0.075 + 0.10 = 0.30 m³ → air = 0.30/0.21 = **1.43 m³/m³**.

Bubbling FBC: sand/limestone bed fluidised by air, 750-950 °C, drying → devolatilisation (volatiles burn in bubbles/freeboard) → char burns in bed; particle convection to immersed tubes + gas convection + radiation.

Carburetor and mixture

Venturi: V↑ p↓; float chamber at p_{atm}; Δp pushes fuel from jet; throttle after venturi; choke before. Simple carb richens with flow → **compensating jet** (leans with flow) / air-bleed emulsion tube keep A/F constant. Accelerator pump for transients.

Requirements: idle rich (~12, dilution by residual), cruise lean (best economy), full power rich (~12.5), cold start very rich, acceleration extra squirt.

Diluting: more expansion work (γ↑), less pumping, less heat loss and NOx.

Venturi numerical

$C_2 = C_v\sqrt{[2c_pT_1(1 - (p_2/p_1)^{0.286})]}$ → p_2; $\rho_1 = p_1/RT_1$; $\rho_2 = \rho_1(p_2/p_1)^{0.714}$; $A = \dot{m}/\rho_2C_2$; $d = \sqrt{(4A/\pi)}$.

2024: 0.1 kg/s, 1.02 bar, 298 K, 100 m/s, C_v 0.75 → p_2 = 0.918 bar, ρ_2 = 1.106, A = 9.04 cm², **d = 33.9 mm** (slide's simplified way: 37.8 mm).

2023 (take 102 m/s): 0.0833 kg/s, 1.05 bar, C_v 0.8 → **d = 30.1 mm**.

Fuel injection

Purpose: meter, time, rate, atomize, distribute, sharp cut-off. Parts: tank, feed pump, filters, HP pump, governor, HP lines, injector. Air injection (old) vs solid: individual pump, unit injector, distributor, common rail (1000-2000 bar, ECU). SI: TBI, MPFI, GDI.

Nozzles: single-hole (15°, dribbles), multi-hole (4-18 × 0.15-0.35 mm, DI), pintle (hollow cone 0-60°, low pressure, IDI), pintaux (auxiliary hole, cold start). **Sac volume** → late HC; VCO nozzle.

Cooling and lubrication

Need cooling: 2500 °C gas, oil film, seizure, thermal stress, knock, η_v. Overcooling: poor vaporization, oil dilution, corrosion, friction, low η.

Air (fins; bikes, aircraft, small) vs liquid (jackets, radiator, pump, fan, thermostat; cars, large; uniform). Circuit: jacket → thermostat → top tank → core + fan → bottom tank → pump. Thermostat wax/bellows, opens ~70-80 °C; stuck shut = overheat, stuck open = overcool. Thermosyphon: density difference, slow, stops if level low.

Lube: friction, cooling, sealing, cleaning, shock. SAE W at −18 °C, summer at 99 °C, multigrade 20W-50. Mist (petroil) 1:50 in 2-stroke: no pump, but smoke/deposits. Wet sump (splash, pressure), dry sump.

Pollutants

SI HC: crevices (38 %), quench, oil film, deposits, liquid fuel. CO: rich. NOx: Zeldovich O + N_2 → NO + N, N + O_2 → NO + O, N + OH → NO + H; > 1800 K, peaks slightly lean, grows with time. Also prompt (Fenimore) and fuel-N.

CI: over-lean spray edge HC, sac HC, soot in rich core, NOx in diffusion flame (NO_2 10-30 %), low CO.

Smog: NO_2 + hv → NO + O → O_3, PAN.

PM: filter-collected at ≤ 325 K: soot + organic fraction. Measure: dilution tunnel + filter, Bosch smoke, Hartridge opacity. CI >> SI (heterogeneous). Smoke: black = soot, blue = oil, white = fuel/water.

Catalytic converter and control

Steel shell, cordierite honeycomb, Al_2O_3 wash-coat, Pt/Pd (oxidise CO, HC), Rh (reduce NO), CeO_2 (O_2 store). 2CO + O_2 → 2CO_2; HC + O_2 → CO_2 + H_2O; 2NO + 2CO → N_2 + 2CO_2.

Light-off 250-300 °C (S-curve); > 800-900 °C sinters. φ window at 1: lean → NOx conversion drops; rich → CO/HC drop. Lead poisons.

Control: engine design/operation, after-treatment (thermal reactor > 600 °C, catalyst, DPF, SCR), fuel change, PCV, canister. NOx: EGR, retard, water injection, lean, SCR. Blow-by: gas past rings → PCV valve to intake.

Section B: performance, cycles, combustion, knock, supercharging

Performance

$\eta_v = 2\dot{m}_a/(\rho_a V_d N)$; power $\propto \eta_v$. BSFC = $\dot{m}_f$/BP. IMEP = as/l. BP = $2\pi NT/60$. FP = IP − BP; η_m = BP/IP; η_{bth} = $3600BP/(\dot{m}_f CV)$.

Curves vs speed: η_v and torque hump (max mid speed); BP peaks at higher speed; BSFC bowl (min near max-torque speed).

Mean piston speed 2LN = 8-15 m/s: above it choking, inertia, friction; below it low power.

Late IVC: ram effect at high speed, backflow at low speed. Quasi-static: fuel vapour partial pressure, p_e/p_i and r (residual).

Rope brake (2024 6b)

BP = $(W - S)\pi DN/60 = 350\pi(0.6)(600)/60$ = **6.60 kW**. p_{mi} = 420 × 1.1/60 = **7.7 bar**. IP = $p_{mi}LAn/60$, n = N/2 = 300 → **4.54 kW**. η_m = 1.45 (data inconsistent; say so). η_{bth} = 6.597 × 3600/(0.815 × 44 000) = 66.2 %. BSFC = 0.124 kg/kWh.

Pumping and lean CI

Upper loop = gross indicated work; lower loop = pumping work; net = A − B. SI throttle → low intake p → big pumping loss at part load; CI no throttle. Reduce: VVT/early IVC, lean/EGR, GDI stratified, downsizing, cylinder deactivation. WOT: throttle wide open, max torque, min pumping. Turbo: $p_{in} > p_{ex}$ → pumping work positive.

CI lean: quality governed (no throttle, full air), spray must find O_2, smoke limit φ ~0.7-0.8.

Valve timing

IVO 10-30° bTDC; IVC 30-60° aBDC (ram); EVO 30-55° bBDC (blow-down); EVC 10-20° aTDC (scavenging); spark 20-30° bTDC. Reasons: valves open gradually, gas inertia, blow-down, finite burn. Overlap: good scavenging at speed, backflow and charge loss at low speed.

Inlet valve larger: intake Δp only ~0.1-0.2 bar vs exhaust blow-down 3-5 bar; smaller exhaust valve easier to cool.

Cycles

Otto: $T_2/T_1 = T_3/T_4 = r^{\gamma-1}$ → **$\eta = 1 - 1/r^{\gamma-1}$** (r = 8: 56.5 %).

Diesel: $\eta = 1 - (\rho^{\gamma} - 1)/[\gamma\, r^{\gamma-1}(\rho - 1)]$; ρ↑ → η↓. γ↑ → η↑. Same r: Otto > Dual > Diesel; same p_{max}: Diesel > Dual > Otto.

2-stroke and scavenging

Scavenging = fresh charge pushes exhaust out (cross, loop, uniflow). Critical in 2-stroke (all exchange in ~130° around BDC, both ports open). 2-stroke SI scavenges with mixture → 20-30 % short-circuit, HC, oil smoke → limited use. 2-stroke CI scavenges with air → no fuel loss → large marine engines. 4-stroke: better economy, less pollution.

HCCI, DISI, SI mixture formation

Mixture: carburetor, port injection (TBI/MPFI), direct injection. HCCI: homogeneous lean charge auto-ignites; low NOx and soot, high η; timing hard, narrow load, HC/CO. DISI: in-cylinder petrol injection; charge cooling, higher r, stratified lean unthrottled; PM, NOx after-treatment.

SI and CI combustion

SI: I ignition lag A-B (~10-20°, chemical, 5-10 % burnt), II flame propagation B-C (80-90 %, 10-30 m/s, peak 10-15° aTDC), III after-burning C-D.

CI (ROHR): ignition delay → premixed spike → mixing-controlled hump → late tail.

SIT = lowest temperature for self-ignition. Premixed (SI, blue, kinetics) vs diffusion (CI, yellow soot, mixing).

Abnormal combustion

Surface ignition: hot plug, exhaust valve, deposits. Pre-ignition (before spark): uncontrolled advance, negative work, run-away, holed piston; post-ignition (after spark): only faster burn. End gas: unburnt charge ahead of flame. Flame front: thin reaction zone.

SI knock: end gas auto-ignites at END of combustion if $t_{flame} > t_{delay}$; pressure waves 5 kHz. CI knock: long delay → violent premixed burn at START. Factors opposite (SIT, delay, r, T_{in}, p_{in}, wall T, speed, size). SI knock-limited (r, boost, timing); CI not (smoke and stress limited).

CI fuel statements

CI delay down with: high CN, high r, high T_{in}/p_{in}, load, swirl, fine spray, optimum timing, pilot injection.

Good SI fuel (high ON, high SIT, long delay) = bad CI fuel (needs high CN, short delay).

Large slow CI: delay in ms is few crank degrees; big hot cylinders → can burn low-cetane heavy oil.

Smoke limited: more fuel → rich spray core can't find O_2 → black smoke caps max fuel and power.

Supercharging

Supercharger (mechanical: Roots, vane, centrifugal) vs turbocharger (exhaust turbine). Effects: power ↑ 30-50 %, η_m↑, positive pumping work, better combustion; thermal/mechanical load, SI knock, NOx.

CI more suitable (boost shortens delay, less knock, pushes smoke limit; air only). SI limited by knock.

Modifications: lower r, stronger parts, more overlap (up to 60°), cooling, lubrication, intercooler, bigger pump/nozzles, retarded timing, turbine manifold, waste-gate. Turbo limits: special manifold, cam overload, turbine sensitive to gas velocity, lag/range.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
A/F	air-fuel ratio
API	American Petroleum Institute (gravity scale)
BDC	bottom dead centre
BMEP	brake mean effective pressure
BP	brake power
BSFC	brake specific fuel consumption
CFR	Cooperative Fuel Research (standard test engine)
CI	compression ignition (diesel engine)
CN	cetane number
CNG	compressed natural gas
CO	carbon monoxide
CV	calorific value
DISI	direct injection spark ignition
DPF	diesel particulate filter
EGR	exhaust gas recirculation
EVC	exhaust valve closes
EVO	exhaust valve opens
F/A	fuel-air ratio
FAME	fatty acid methyl ester (biodiesel)
FBC	fluidised bed combustion
FC	fixed carbon
FP	friction power
GDI	gasoline direct injection
HC	hydrocarbons (unburnt fuel)
HCCI	homogeneous charge compression ignition
HCV	higher calorific value
IC	internal combustion
IMEP	indicated mean effective pressure
IP	indicated power
ISFC	indicated specific fuel consumption
IVC	inlet valve closes
IVO	inlet valve opens
LCV	lower calorific value
LPG	liquefied petroleum gas
MEP	mean effective pressure
NG	natural gas

NOx	nitrogen oxides (NO + NO_2)
ON	octane number
PCV	positive crankcase ventilation
PM	particulate matter
ROHR	rate of heat release
rpm	revolutions per minute
RVP	Reid vapour pressure
SAE	Society of Automotive Engineers (oil grade)
SCR	selective catalytic reduction
SI	spark ignition (petrol engine)
SIT	self-ignition temperature
SOF	soluble organic fraction
TDC	top dead centre
TWC	three-way catalytic converter
VM	volatile matter
WOT	wide open throttle